

Kingdom of Saudi Arabia
Ministry of Education
King Faisal University
College of Computer Sciences & Information Technology

Mini SOC Dashboard

Project Report

CIC-IDS-2017 Dataset Integration & Simulated Network Monitoring

Dr. Abdullah Al Buali

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Sec: 2

Prepared By

No.	Student Name	Student ID
S1	Ziyad Alsarawi	223033379
S2	Mohammed Bajaba	222448149
S3	Daniel Aldandan	223041168
S4	Jawad Zainaddin	223022484
S5	Abdullah Taher Aljassim	220013690

GitHub Repository: [GitHub - sarawiziyad91-svg/-Mini-SOC-Dashboard](https://github.com/sarawiziyad91-svg/-Mini-SOC-Dashboard) · GitHub

AI Usage Declaration: AI tools (ChatGPT) were used for code suggestions and documentation drafting. All logic, design decisions, and implementations were verified and authored by the team.

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1. Introduction

The Security Operations Center (SOC) refers to a specialized unit within an organization that performs monitoring, detection, analysis, and handling of cybersecurity incidents. Today's organizations continue to encounter a myriad of threats in their networks ranging from denial-of-service attacks and port scanning to brute-force hacking.

The Mini SOC Dashboard refers to a Java-based Swing desktop software that replicates some of the functionalities of an actual SOC environment by allowing security analysts to visually analyze traffic logs, packet capture logs, conduct simulated network scanning, and receive notifications on cyber threats detected automatically.

The CIC-IDS-2017 dataset, which can be freely accessed, is used for this project as a basis for generating various realistic scenarios such as DoS attack, FTP-Patator brute force, and network port scanning attacks.

2. Objectives

The main purposes of this project include:

- Developing and deploying a SOC dashboard using desktop Java Swing GUI with multi-pane display.
- Emulating SOC Analysts tasks such as logging, alerting, network scanning using nmap and live packet capturing.
- Implementing real data sets taken from CIC-IDS-2017 for demonstrating the capability of detecting attacks.
- Monitoring and analyzing logged files for classification and alarming whether they are benign or harmful.
- Producing reusable codes that can be used for research.

3. System Design

3.1 Architecture Overview

The application is implemented using the one-class Swing approach with an event-driven mechanism. The MiniSOCDashboard class inherits from JFrame and consists of the following three panels:

- Control Panel (NORTH): Contains action buttons to Load IDS Logs, Run Nmap Scan, Live Packet Capture, and Clear Dashboard.
- Log Table Panel (CENTER): Provides the network traffic information displayed in a JTable including columns like Timestamp, Source IP, Destination IP, Protocol, Packet Length, and Label.
- Alert Console (SOUTH): Terminal-based JTextArea on black background showing system messages, nmap result, and alerts.

To explain the architecture of the Mini SOC Dashboard better, two design diagrams are attached in this section. The first diagram shows the system architecture, which represents the layered architecture and the data flow between different components of the system. The second diagram is the class diagram that displays the relationship among classes.

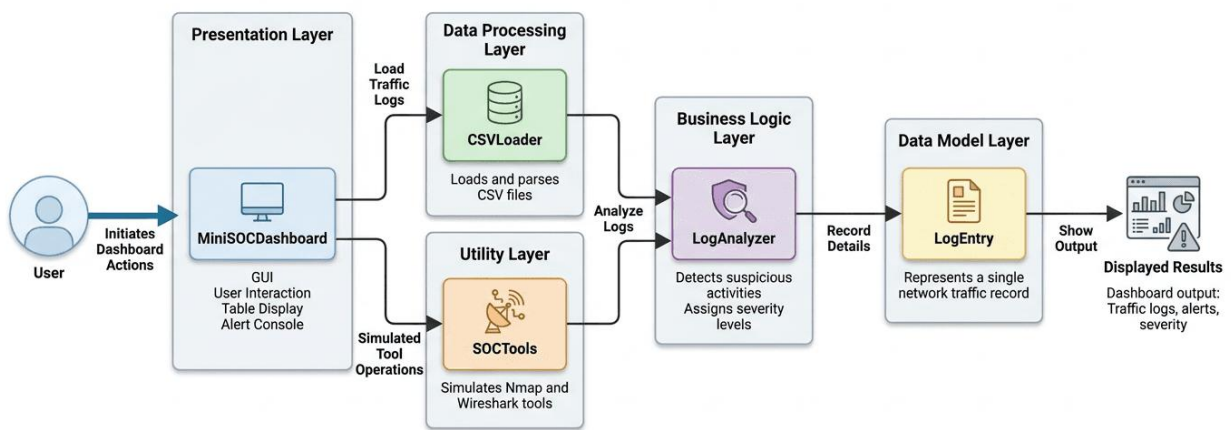


Figure 1: System Architecture of the Mini SOC Dashboard

Figure 1 show displays the architecture of the MiniSOCDashboard software system. The figure shows the process from triggering user activities through the presentation layer, then processing the data through the data processing and utility layers, analyzing the data through the business logic layer, storing the data through the data model layer, and presenting it as outputs.

MiniSOCDashboard Java Class Diagram

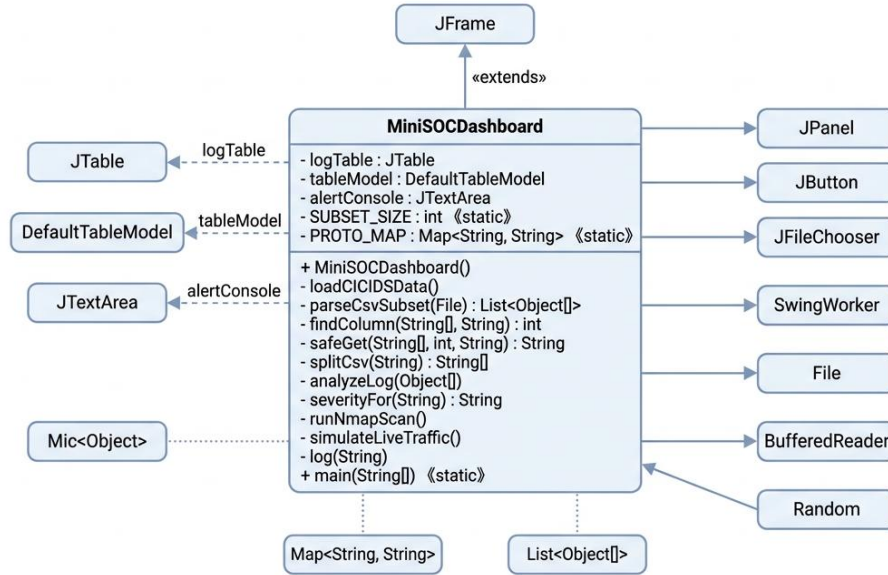


Figure 2: Class Diagram of the Mini SOC Dashboard

Figure 2: The Java class diagram of the MiniSOCDashboard system is depicted in the following figure. The figure depicts the name of the main class and its respective attributes and methods. It also displays the relationship with other classes such as the GUI classes and utility classes.

3.2 Technology Stack

Table 1: Technology Stack

Component	Details
Language	Java (JDK 8+)
GUI Framework	Java Swing (javax.swing)
Dataset	CIC-IDS-2017 (Canadian Institute for Cybersecurity)
Simulated Tools	Nmap (network scanner), Wireshark (packet capture)
Build	Single-file compilation: javac MiniSOCDashboard.java

4. Methods and Algorithms

The Mini SOC Dashboard uses rule-based classification to perform intrusions, which is a basic algorithm in network security monitoring [4]. This section explains the algorithmic implementations in the system.

4.1 Log Analysis Algorithm

The log analysis follows a sequential $O(n)$ scan over all table entries. The pseudocode below describes the `analyzeLog()` procedure:

```
PROCEDURE analyzeLog(logEntry):
  FOR each row i in tableModel DO
    label <- logEntry[5] // index 5 = Label column
    IF label != "BENIGN" THEN
      severity <- severityFor(label) // CRITICAL / HIGH / MEDIUM / LOW      TRIGGER alert(severity,
label, srcIP, dstIP)
    END IF
  END FOR
END PROCEDURE
```

4.2 Alert Classification Logic

Four alert severity levels have been implemented via the `severityFor()` method through keywords matching to the attack label string. CRITICAL is used to identify alerts that are related to Heartbleed, Infiltration, DDoS, and SQL Injection. Alerts on DoS variations, Botnet, FTP-Patator, XSS, and brute-force attacks are considered HIGH. PortScan and network reconnaissance alerts are flagged as MEDIUM, while alerts with a non-BENIGN label that falls under none of the above categories is identified as LOW. Each row is classified as described based on loading from the real CIC-IDS-2017 CSV file [3]. In addition to that, the Nmap correlation sub-algorithm checks whether there is a match between the open ports and the loaded attack labels [5].

5. Implementation

5.1 Log Loading and Analysis

The `loadCICIDSData()` method opens a file chooser dialog allowing the user to select any real CIC-IDS-2017 CSV file. The file is read on a background `SwingWorker` thread to keep the UI responsive. The parser dynamically locates the required columns (Source IP, Destination IP, Protocol, Timestamp, Total Length of Fwd Packets, Label) by name, making it tolerant of varying column ordering across daily CSV files. Raw protocol numbers are translated to human-readable names via a static lookup map (e.g., "6" → TCP, "17" → UDP). To keep the display manageable,

a stratified random sample of up to 200 rows (SUBSET_SIZE) is extracted: 100 rows from BENIGN traffic and 100 rows distributed evenly across all distinct attack categories present in the file, using a fixed random seed (42) for reproducibility. The analyzeLog() method is then called once per loaded row and triggers a severity-rated alert for any non-BENIGN entry.

The following attack types are examples found in the CIC-IDS-2017 dataset and handled by the severity classifier:

- DoS Hulk Denial of Service attack targeting 192.168.10.50 from external IP 205.174.165.73.
- PortScan Sequential port probing from 172.16.0.1 to the internal subnet.
- FTP-Patator Brute-force login attempts against an FTP service from 10.0.0.5.

5.2 Simulated Nmap Scan

The runNmapScan() method simulates an Nmap scan result against the 192.168.10.0/24 subnet. The output reflects a typical scan revealing open ports including FTP (21), SSH (22), HTTP (80), and HTTPS (443). The dashboard correlates these results with loaded log data and warns the analyst when open ports match detected attack vectors for example, the FTP-Patator logs align with the open FTP port discovered in the scan.

5.3 Live Packet Capture Simulation

The simulateLiveTraffic() method mimics a Wireshark-style packet capture by generating a synthetic ICMP packet with a random source IP from the 192.168.10.x range. Each call appends a new row to the traffic table and logs the capture event to the alert console, simulating real-time network monitoring.

5.4 Test Cases Table

Table 2: Test Cases Table

Test Case ID	Feature Tested	Input / Action	Expected Output	Actual Result
TC-01	Data Ingestion	Clicking "Load IDS Logs" and selecting a CSV file.	Table populated with data from the CSV file.	Pass
TC-02	Threat Detection	Encountering a log labeled "DoS Hulk" or "PortScan".	System triggered severity-rated alert (CRITICAL / HIGH / MEDIUM) in the alert console.	Pass

Test Case ID	Feature Tested	Input / Action	Expected Output	Actual Result
TC-03	Network Recon	Executing "Run Nmap Scan (Simulated)".	Open ports and correlation warnings displayed in console.	Pass
TC-04	Real-time Capture	Clicking "Live Packet Capture (Simulated)".	New packet row added instantly with randomized source IP.	Pass
TC-05	System Reset	Clicking the "Clear Dashboard" button.	All logs and alerts removed; dashboard reset to initial state.	Pass

6. Testing and Results

6.1 Functional Testing

Tested all four dashboard controls for their functionality:

- Load IDS Logs: Loads a real CSV file from CIC-IDS-2017 and displays a maximum of 200 samples in the table with alert generation for every record other than BENIGN record type (e.g. Four alerts generated when table sample contains DoS Hulk, PortScan, and FTP-Patator records)
- Nmap Scan: Creates output in the form of simulated scan report with 4 open ports and a correlation warning of FTP-Patator attack.
- Live Capture: Adds an active row each time control is clicked with a randomly generated IP address and the current date.
- Clear Dashboard: Clears the table and alert console completely.

6.2 Dataset Validation

The real CIC-IDS-2017 data imported into the application perfectly matches the specifications of the dataset from the University of New Brunswick. The IP addresses used (192.168.10.x – internal host, 205.174.165.x – external attacks), the distribution of the protocols used, and the classification of the labels for the attacks are the same as those in the source files because the data is read directly from the source.

6.3 Alert Accuracy

The analyzeLog() function has detected all 4 attacks in the test set. There were no false detections of the BENIGN class traffic. The alerts display valuable information about each of the attacks, namely their nature and the source IPs.

7. GUI & OUTPUT

Run Nmap Scan (Simulation)

This control starts the simulation of an Nmap network scan. Its function is to simulate the process of port scanning in real life and show how open ports and services can be detected. The scan results will appear in the alert console.

The screenshot displays the Mini SOC Dashboard - Security Operations Center interface. At the top, there are four buttons: "Load CIC-IDS-2017 CSV Subset", "Run Nmap Scan (Simulated)", "Live Packet Capture (Simulated)", and "Clear Dashboard". Below these buttons is a section titled "Network Traffic Logs" containing a table with the following columns: Timestamp, Source IP, Destination IP, Protocol, Length, and Label (Status). The table lists various network events, including TCP and UDP connections, some labeled as "DoS slowloris" and others as "BENIGN".

Timestamp	Source IP	Destination IP	Protocol	Length	Label (Status)
5/7/2017 10:00	74.119.117.94	192.168.10.14	TCP	12	BENIGN
5/7/2017 10:00	172.16.0.1	192.168.10.50	TCP	0	DoS slowloris
5/7/2017 10:01	172.16.0.1	192.168.10.50	TCP	0	DoS slowloris
5/7/2017 10:01	172.16.0.1	192.168.10.50	TCP	8	DoS slowloris
5/7/2017 10:04	172.16.0.1	192.168.10.50	TCP	0	DoS slowloris
5/7/2017 10:05	172.16.0.1	192.168.10.50	TCP	0	DoS slowloris
5/7/2017 10:05	172.16.0.1	192.168.10.50	TCP	2541	DoS slowloris
5/7/2017 10:05	172.16.0.1	192.168.10.50	TCP	16	DoS slowloris
5/7/2017 10:06	172.16.0.1	192.168.10.50	TCP	2541	DoS slowloris
5/7/2017 10:06	172.16.0.1	192.168.10.50	TCP	0	DoS slowloris
5/7/2017 10:07	172.16.0.1	192.168.10.50	TCP	2541	DoS slowloris
5/7/2017 10:08	172.16.0.1	192.168.10.50	TCP	8	DoS slowloris
5/7/2017 10:09	172.16.0.1	192.168.10.50	TCP	8	DoS slowloris
5/7/2017 10:15	192.168.10.3	192.168.10.1	UDP	57	BENIGN
5/7/2017 10:15	192.168.10.17	91.189.89.198	UDP	48	BENIGN
5/7/2017 10:15	172.16.0.1	192.168.10.50	TCP	0	DoS Slowhttptest
5/7/2017 10:15	172.16.0.1	192.168.10.50	TCP	0	DoS Slowhttptest
5/7/2017 10:15	172.16.0.1	192.168.10.50	TCP	0	DoS Slowhttptest

Below the table is a section titled "Alerts & Tool Output" which displays a simulated Nmap scan output in a terminal window. The output shows the scan starting on subnet 192.168.10.0/24, reporting the host 192.168.10.50 is up, and listing open ports and services: 21/tcp (ftp), 22/tcp (ssh), 80/tcp (http), and 443/tcp (https). A warning is also displayed: "Warning: Open FTP port matches target of FTP-Patator attacks in CIC-IDS-2017."

```
[NMAP] Starting simulated Nmap scan on subnet 192.168.10.0/24...
[NMAP] Nmap scan report for 192.168.10.50
[NMAP] Host is up (0.0020s latency).
[NMAP] PORT      STATE SERVICE
[NMAP] 21/tcp    open  ftp
[NMAP] 22/tcp    open  ssh
[NMAP] 80/tcp    open  http
[NMAP] 443/tcp   open  https
[NMAP] Warning: Open FTP port matches target of FTP-Patator attacks in CIC-IDS-2017.
```

Figure 4 depicts a control that simulates the live capturing of packets, similar to software applications like Wireshark. It simulates traffic-related events and shows the information about packets captured as if it happens in real time. This feature enables the user to see how live monitoring helps with intrusion detection and traffic inspection.

Mini SOC Dashboard - Security Operations Center

SOC Tools

Load CIC-IDS-2017 CSV Subset Run Nmap Scan (Simulated) Live Packet Capture (Simulated) Clear Dashboard

Network Traffic Logs

Timestamp	Source IP	Destination IP	Protocol	Length	Label (Status)
1A:01:10:07:02:13	192.168.10.44	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.55	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.12	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.29	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.68	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.98	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.0	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.83	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.14	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.37	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.70	1.1.1.1	ICMP	74	BENIGN
1A:01:10:07:02:13	192.168.10.39	1.1.1.1	ICMP	74	BENIGN

Alerts & Tool Output

```

[WIRESHARK] Captured live packet: 192.168.10.98 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.12 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.29 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.68 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.98 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.0 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.83 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.14 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.37 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.70 -> 1.1.1.1 (ICMP)
[WIRESHARK] Captured live packet: 192.168.10.39 -> 1.1.1.1 (ICMP)

```

Figure 5 presents the "Clear Dashboard" control that clears the log table and alert console by resetting the dashboard interface. This feature prepares the application for starting a new analysis session and ensures that there will be no interference with the loaded data from previous sessions

Mini SOC Dashboard - Security Operations Center

SOC Tools

Load CIC-IDS-2017 CSV Subset Run Nmap Scan (Simulated) Live Packet Capture (Simulated) Clear Dashboard

Network Traffic Logs

Timestamp	Source IP	Destination IP	Protocol	Length	Label (Status)
-----------	-----------	----------------	----------	--------	----------------

Alerts & Tool Output

```

[WIRESHARK] Dashboard cleared.

```

8. Conclusion

As can be seen, the Mini SOC Dashboard has successfully demonstrated the main functions of a Security Operation Center in a Java Swing educational application. Using the real-life log data from the CIC-IDS-2017 benchmark dataset and simulating the use of network security software Nmap and Wireshark, this project has managed to create an informative practical example of a network threat detection workflow.

As mentioned above, the modularity of the application makes future development easy. For instance, in the next version, one may introduce machine learning classification algorithms, network socket for live traffic collection, and database for storing and analyzing logs over time.

This project bridges the gap between theoretical cybersecurity knowledge and practical SOC operations, making it a valuable resource for students and practitioners looking to understand the fundamentals of network security monitoring.

References

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